INCH-POUND

MIL-C-24643/51C
14 March 1994
SUPERSEDING
MIL-C-24643/51B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, MULTICONDUCTOR, TYPE LSMNW (INCLUDING VARIATION LSMNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSMNW (unarmored) and LSMNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated, AWG 18, class B stranding.

Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Cross-linked polyethylene insulation or ethylene propylene rubber insulation., 0.020 inch minimum average thickness. Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.

Fourth - The required number of conductors (see table I), cabled together with a lay not greater than 24 times the pitch diameter of the layer, except that the lay of the two outer layers for size 44 shall not be greater than 18 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to form a firm, well-rounded assembly.

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Construction (non-watertight) (continued)

Fifth - Binder tape over each pair applied helically with overlap.

Construction variant for type LSMNW

* Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.

Construction variant for type LSMNWA

* Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.

* Seventh- Braided metal armor.

TABLE I. Details

Military 1/ part no. M24643/51	Size 2/	Number of conductors	Overall diameter (inches) TYPE LSMNW min	TYPE LSMNW max	TYPE LSMNWA min	TYPE LSMNWA max	Cold bending mandrel diameter (max inch)
-01UN	7	7	0.370	0.400	0.420	0.450	3
-02UN	10	10	.457	.495	.507	.545	3
-03UN	14	14	.494	.535	.544	.585	4
-04UN	19	19	.545	.590	.595	.640	4
-05UN	24	24	.633	.685	.683	.735	5
-06UN	30	30	.670	.725	.720	.775	5
-07UN	37	37	.726	.785	.776	.835	6
-08UN	44	44	.823	.890	.873	.940	7

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSMNW and LSMNWA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.07
Voltage withstand - volts, root mean square, minimum	3000
* Insulation resistance - Conductor to conductor, Megohms/1000 feet, minimum	1000
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
* Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50

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EXAMINATION AND TESTS (continued):

	Requirements
Cold bending, cable - (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Ethylene propylene	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (conductor) - method 1 only, cycles, minimum	25
* Permanence of printing (jacket) - (type LSMNW only) cycles minimum	125
* Armor - Type LSMNWA only, conformance to material, construction and coverage	No failure
Group D:	
* Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:

Type and size	Feet (nominal)
LSMNW-7	1500
LSMNW-10	1500
All other sizes	1000

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government *assumes no* liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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